

Trabalho 02 - Resultados Computacionais Comparativos

Machine Scheduling, Job Shop Scheduling e Flow Shop Scheduling

Item	Valor
Solver	HiGHS
Modelagem	Julia + JuMP
Linguagem	Julia
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1 Parâmetros de execução

Tabela 1: Parâmetros de tempo das execuções

Execução	Machine	Job Shop	Flow Shop
Antiga	1 s	60 s	60 s / 30 s
5 min	300 s	300 s	300 s

2 Resumo geral

Tabela 2: Indicadores globais

Indicador	Valor
Instâncias por execução	48
Soluções ótimas - execução antiga	17
Soluções ótimas - execução 5 min	27
Time limit - execução antiga	31
Time limit - execução 5 min	21
Instâncias com valor melhorado	27
Instâncias com valor igual	21
Instâncias com valor piorado	0
Gap solver médio antigo	48.283%
Gap solver médio 5 min	30.506%
Tempo médio antigo	13.268 s
Tempo médio 5 min	149.800 s

Tabela 3: Resumo da execução antiga

Problema	Inst.	Ótimos	Time limit	Valor médio	Gap solver médio	Gap ótimo médio	Gap heurística médio	Tempo médio	Limites
Flow Shop	7	6	1	220.571	0.639	0.000	1.325	11.676	30, 60
Job Shop	10	3	7	835.700	21.043	4.489	12.837	52.803	60
Machine	31	8	23	143.871	67.829	0.000	23.264	0.875	1

Tabela 4: Resumo da execução com limite de 5 minutos

Problema	Inst.	Ótimos	Time limit	Valor médio	Gap solver médio	Gap ótimo médio	Gap heurística médio	Tempo médio	Limites
Flow Shop	7	6	1	219.857	0.302	0.000	1.570	45.973	300
Job Shop	10	5	5	816.600	14.900	2.570	14.713	196.273	300
Machine	31	16	15	104.323	42.360	0.000	51.313	158.253	300

3 Comparação por problema

Tabela 5: Comparação agregada entre execução antiga e execução de 5 minutos

Problema	Inst.	Melhorou	Igual	Piorou	Ótimos ant.	Ótimos 5 min	TL ant.	TL 5 min	Delta valor médio	Delta gap solver	Delta tempo médio	Razão tempo
Flow Shop	7	1	6	0	6	6	1	1	-0.714	-0.336	34.296	1.578
Job Shop	10	6	4	0	3	5	7	5	-19.100	-6.143	143.470	3.387
Machine	31	20	11	0	8	16	23	15	-39.548	-25.469	157.379	155.205

4 Validações principais

Tabela 6: Validação das instâncias principais

Problema	Instância	Métrica	Esperado	Obtido 5 min	Status
Machine	inst_book	atraso total	16	16	Validado
Job Shop	ft06	Cmax	55	55	Validado
Flow Shop	problem_3m_10j	Cmax	126	126	Validado

5 Figuras comparativas

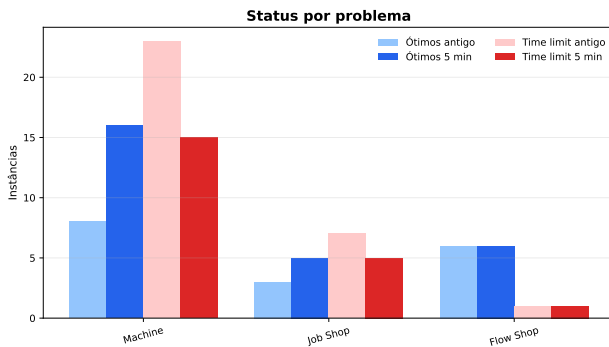


Figura 1: Status por problema

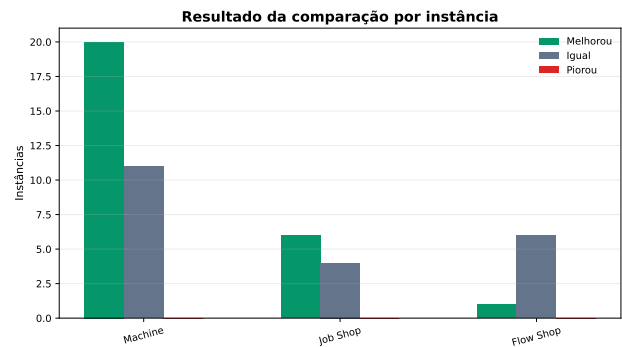


Figura 2: Resultado da comparação por problema

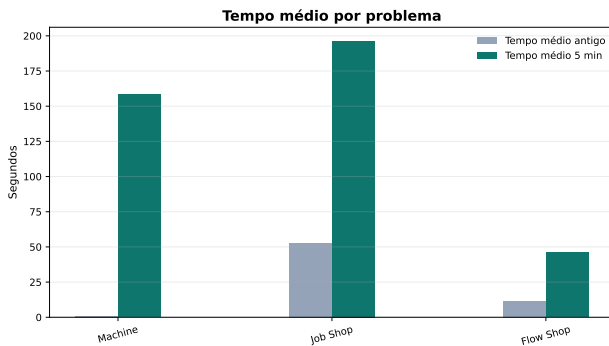


Figura 3: Tempo médio por problema

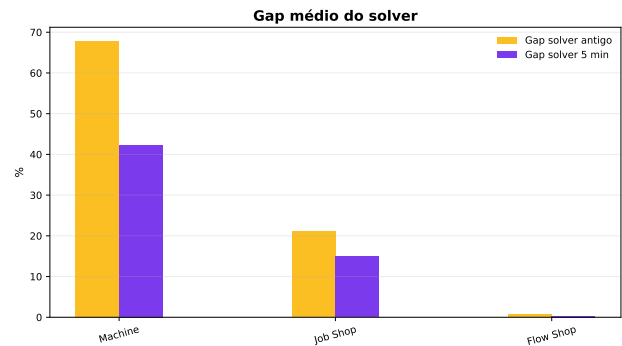


Figura 4: Gap solver médio por problema

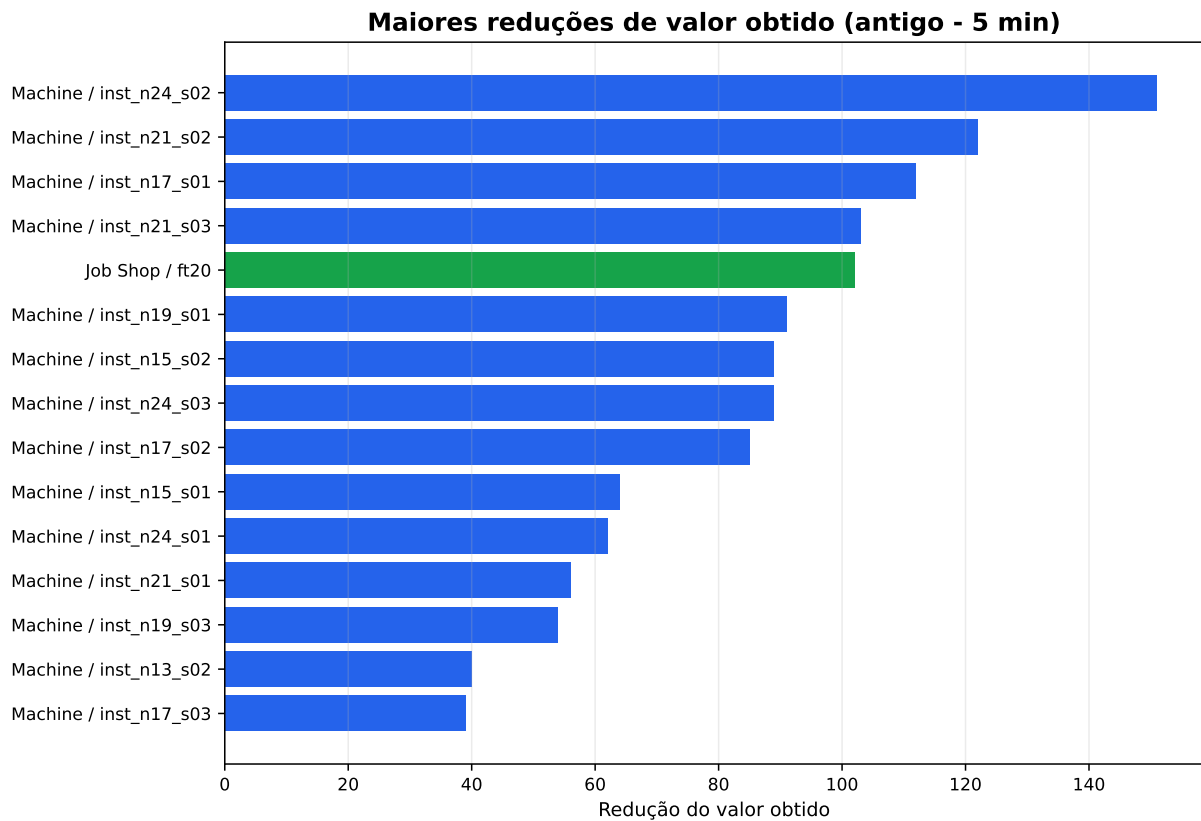


Figura 5: Maiores reduções de valor obtido

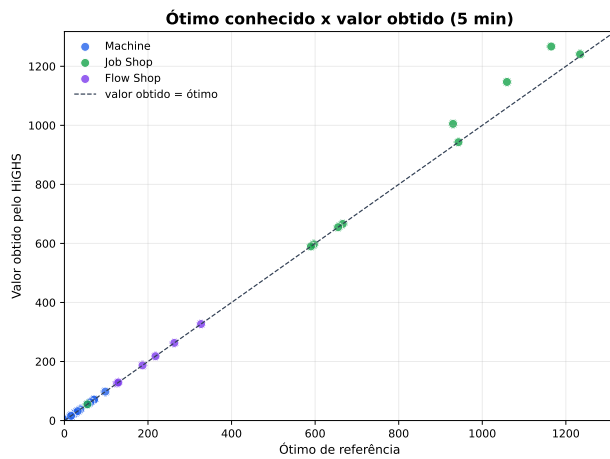


Figura 6: Ótimo conhecido versus valor obtido

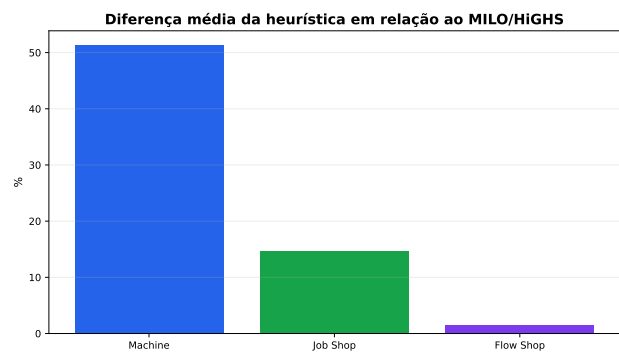


Figura 7: Heurística versus MILO/HiGHS

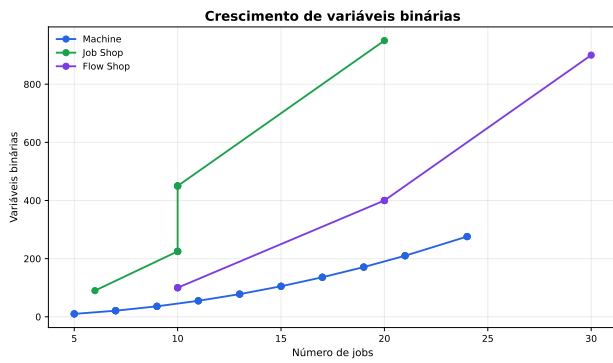


Figura 8: Variáveis binárias por tamanho

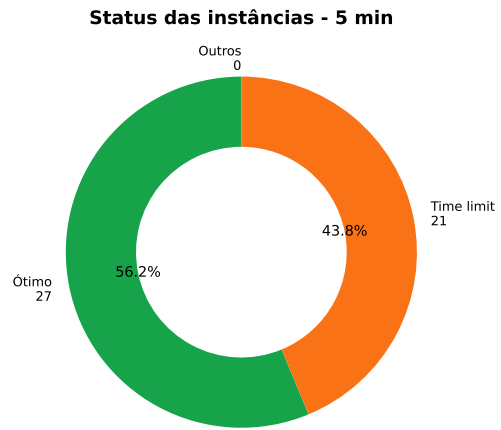


Figura 9: Status na execução de 5 minutos

6 Maiores reduções

Tabela 7: Dez maiores reduções de valor obtido

Problema	Instância	Valor ant.	Valor 5 min	Redução	Redução %
Machine	inst_n24_s02	381	230	151	39.63
Machine	inst_n21_s02	314	192	122	38.85
Machine	inst_n17_s01	280	168	112	40
Machine	inst_n21_s03	282	179	103	36.52
Job Shop	ft20	1369	1267	102	7.45
Machine	inst_n19_s01	340	249	91	26.76
Machine	inst_n15_s02	213	124	89	41.78
Machine	inst_n24_s03	353	264	89	25.21
Machine	inst_n17_s02	243	158	85	34.98
Machine	inst_n15_s01	202	138	64	31.68

7 Instâncias em limite de tempo

Tabela 8: Instâncias que permaneceram com status TIME_LIMIT na execução de 5 minutos

Problema	Instância	Valor 5 min	Gap solver	Tempo
Machine	inst_n11_s03	116	58.141	300.030
Machine	inst_n15_s01	138	77.458	300.029
Machine	inst_n15_s02	124	65.762	300.038
Machine	inst_n17_s01	168	76.593	300.027
Machine	inst_n17_s02	158	91.115	300.041
Machine	inst_n17_s03	108	75.550	300.063
Machine	inst_n19_s01	249	94.465	300.032
Machine	inst_n19_s02	119	97.999	300.030
Machine	inst_n19_s03	237	94.589	300.032
Machine	inst_n21_s01	190	92.034	300.035
Machine	inst_n21_s02	192	99.479	300.047
Machine	inst_n21_s03	179	93.938	300.042
Machine	inst_n24_s01	133	100	300.046
Machine	inst_n24_s02	230	98.815	300.048
Machine	inst_n24_s03	264	97.183	300.050
Job Shop	abz5	1241	15.833	300.425
Job Shop	ft10	1005	25.871	300.010
Job Shop	ft20	1267	61.906	300.095
Job Shop	la03	597	13.902	300.009
Job Shop	orb01	1147	31.474	300.023
Flow Shop	problem_10m_20j	289	2.117	300.031

8 Gráficos de Gantt

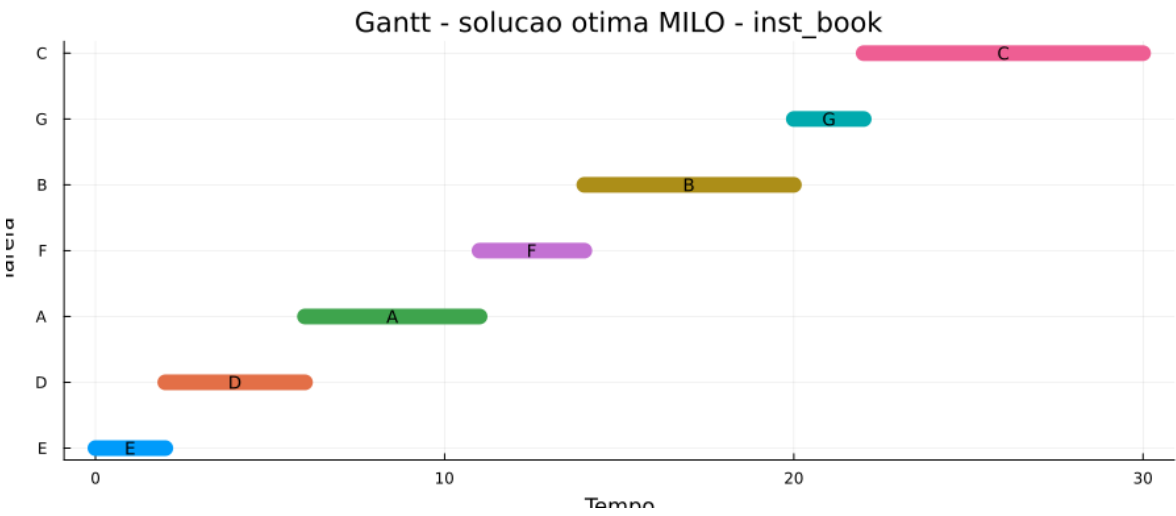


Figura 10: Machine Scheduling - instância inst_book

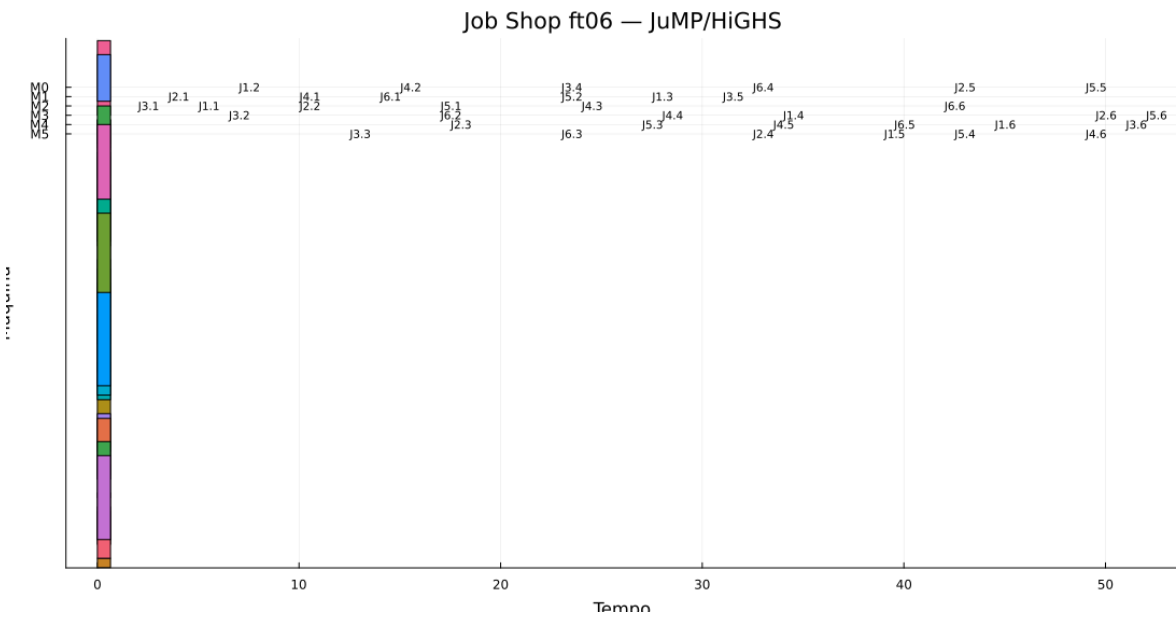


Figura 11: Job Shop Scheduling - instância ft06

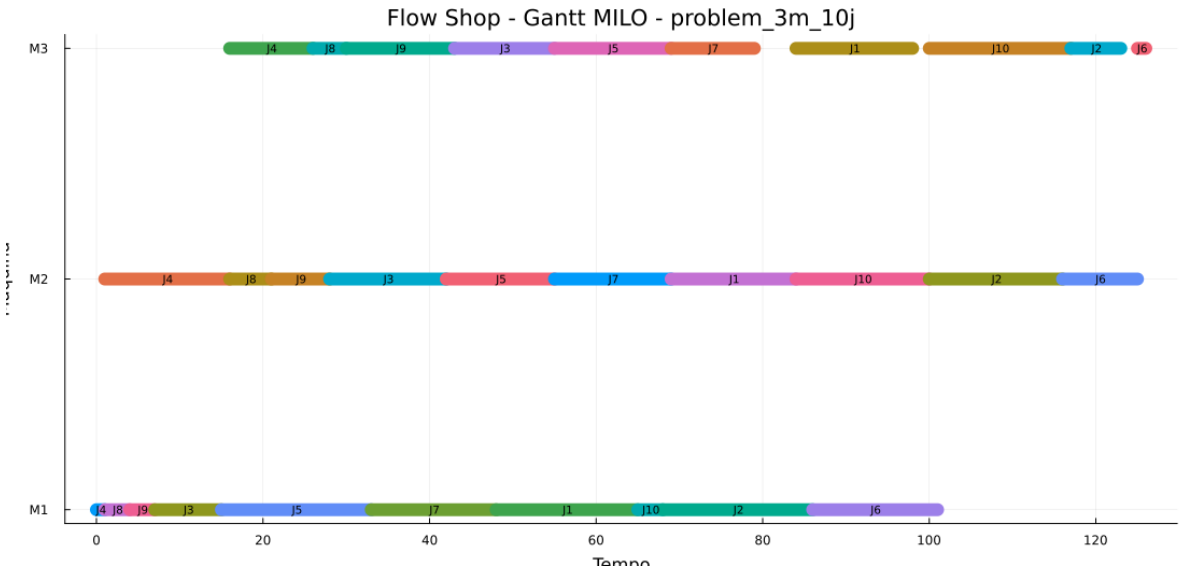


Figura 12: Flow Shop Scheduling - instância problem_3m_10j

9 Comparação por instância

9.1 Valor e status

Problema	Instância	Lim. ant.	Lim. 5 min	Status ant.	Status 5 min	Valor ant.	Valor 5 min	Delta	Delta %	Resultado
Machine	inst_book	1	300	OPT	OPT	16	16	0	-0.00	Igual
Machine	inst_n05_s01	1	300	OPT	OPT	7	7	0	0.00	Igual
Machine	inst_n05_s02	1	300	OPT	OPT	15	15	0	-0.00	Igual
Machine	inst_n05_s03	1	300	OPT	OPT	3	3	0	0.00	Igual
Machine	inst_n07_s01	1	300	OPT	OPT	38	38	0	-0.00	Igual
Machine	inst_n07_s02	1	300	OPT	OPT	30	30	0	-0.00	Igual
Machine	inst_n07_s03	1	300	OPT	OPT	26	26	0	0.00	Igual
Machine	inst_n09_s01	1	300	OPT	OPT	26	26	0	-0.00	Igual
Machine	inst_n09_s02	1	300	TL	OPT	56	56	0	0.00	Igual
Machine	inst_n09_s03	1	300	TL	OPT	32	32	-0	-0.00	Igual
Machine	inst_n11_s01	1	300	TL	OPT	16	16	0	-0.00	Igual
Machine	inst_n11_s02	1	300	TL	OPT	66	62	-4	-6.06	Melhorou
Machine	inst_n11_s03	1	300	TL	TL	120	116	-4	-3.33	Melhorou
Machine	inst_n13_s01	1	300	TL	OPT	92	71	-21	-22.83	Melhorou
Machine	inst_n13_s02	1	300	TL	OPT	138	98	-40	-28.99	Melhorou
Machine	inst_n13_s03	1	300	TL	OPT	73	71	-2	-2.74	Melhorou
Machine	inst_n15_s01	1	300	TL	TL	202	138	-64	-31.68	Melhorou
Machine	inst_n15_s02	1	300	TL	TL	213	124	-89	-41.78	Melhorou
Machine	inst_n15_s03	1	300	TL	OPT	84	62	-22	-26.19	Melhorou
Machine	inst_n17_s01	1	300	TL	TL	280	168	-112	-40	Melhorou
Machine	inst_n17_s02	1	300	TL	TL	243	158	-85	-34.98	Melhorou
Machine	inst_n17_s03	1	300	TL	TL	147	108	-39	-26.53	Melhorou
Machine	inst_n19_s01	1	300	TL	TL	340	249	-91	-26.76	Melhorou
Machine	inst_n19_s02	1	300	TL	TL	135	119	-16	-11.85	Melhorou
Machine	inst_n19_s03	1	300	TL	TL	291	237	-54	-18.56	Melhorou
Machine	inst_n21_s01	1	300	TL	TL	246	190	-56	-22.76	Melhorou
Machine	inst_n21_s02	1	300	TL	TL	314	192	-122	-38.85	Melhorou
Machine	inst_n21_s03	1	300	TL	TL	282	179	-103	-36.52	Melhorou
Machine	inst_n24_s01	1	300	TL	TL	195	133	-62	-31.79	Melhorou
Machine	inst_n24_s02	1	300	TL	TL	381	230	-151	-39.63	Melhorou
Machine	inst_n24_s03	1	300	TL	TL	353	264	-89	-25.21	Melhorou
Job Shop	abz5	60	300	TL	TL	1250	1241	-9	-0.72	Melhorou
Job Shop	abz6	60	300	OPT	OPT	943	943	0	-0.00	Igual
Job Shop	ft06	60	300	OPT	OPT	55	55	0	0.00	Igual
Job Shop	ft10	60	300	TL	TL	1030	1005	-25	-2.43	Melhorou
Job Shop	ft20	60	300	TL	TL	1369	1267	-102	-7.45	Melhorou
Job Shop	la01	60	300	TL	OPT	666	666	0	-0.00	Igual
Job Shop	la02	60	300	TL	OPT	665	655	-10	-1.50	Melhorou
Job Shop	la03	60	300	TL	TL	614	597	-17	-2.77	Melhorou
Job Shop	la04	60	300	OPT	OPT	590	590	0	0.00	Igual
Job Shop	orb01	60	300	TL	TL	1175	1147	-28	-2.38	Melhorou
Flow Shop	problem_10m_20j	60	300	TL	TL	294	289	-5	-1.70	Melhorou
Flow Shop	problem_3m_10j	60	300	OPT	OPT	126	126	0	0.00	Igual
Flow Shop	problem_3m_20j	60	300	OPT	OPT	218	218	0	0.00	Igual
Flow Shop	problem_5m_10j	60	300	OPT	OPT	129	129	0	0.00	Igual
Flow Shop	problem_5m_20j	60	300	OPT	OPT	263	263	0	0.00	Igual
Flow Shop	problem_5m_30j	30	300	OPT	OPT	327	327	0	0.00	Igual
Flow Shop	problem_8m_10j	60	300	OPT	OPT	187	187	0	0.00	Igual

9.2 Gap

Problema	Instância	Gap solver ant.	Gap solver 5 min	Delta gap solver	Gap ótimo ant.	Gap ótimo 5 min	Delta gap ótimo
Machine	inst_book	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n05_s01	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n05_s02	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n05_s03	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n07_s01	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n07_s02	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n07_s03	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n09_s01	0.000	0.000	0.000	0.000	0.000	0.000
Machine	inst_n09_s02	60.863	0.000	-60.863	0.000	0.000	0.000
Machine	inst_n09_s03	76.014	0.008	-76.006	0.000	0.000	-0.000
Machine	inst_n11_s01	81.250	0.000	-81.250	0.000	0.000	0.000
Machine	inst_n11_s02	83.701	0.009	-83.692	6.452	0.000	-6.452
Machine	inst_n11_s03	94.257	58.141	-36.116	NA	NA	NA
Machine	inst_n13_s01	87.178	0.008	-87.170	29.577	0.000	-29.577
Machine	inst_n13_s02	92.350	0.010	-92.340	40.816	0.000	-40.816
Machine	inst_n13_s03	83.225	0.000	-83.225	2.817	0.000	-2.817
Machine	inst_n15_s01	93.885	77.458	-16.427	NA	NA	NA
Machine	inst_n15_s02	97.455	65.762	-31.693	NA	NA	NA
Machine	inst_n15_s03	81.333	0.000	-81.333	35.484	0.000	-35.484
Machine	inst_n17_s01	95.711	76.593	-19.118	NA	NA	NA
Machine	inst_n17_s02	96.202	91.115	-5.088	NA	NA	NA
Machine	inst_n17_s03	95.332	75.550	-19.782	NA	NA	NA
Machine	inst_n19_s01	96.725	94.465	-2.260	NA	NA	NA
Machine	inst_n19_s02	100	97.999	-2.001	NA	NA	NA
Machine	inst_n19_s03	96.737	94.589	-2.148	NA	NA	NA
Machine	inst_n21_s01	94.920	92.034	-2.885	NA	NA	NA
Machine	inst_n21_s02	100	99.479	-0.521	NA	NA	NA
Machine	inst_n21_s03	97.187	93.938	-3.250	NA	NA	NA
Machine	inst_n24_s01	100	100	0.000	NA	NA	NA
Machine	inst_n24_s02	100	98.815	-1.185	NA	NA	NA
Machine	inst_n24_s03	98.363	97.183	-1.181	NA	NA	NA
Job Shop	abz5	19.675	15.833	-3.842	1.297	0.567	-0.729
Job Shop	abz6	0.000	0.000	0.000	0.000	0.000	0.000
Job Shop	ft06	0.000	0.000	0.000	0.000	0.000	0.000
Job Shop	ft10	27.670	25.871	-1.799	10.753	8.065	-2.688
Job Shop	ft20	65.361	61.906	-3.455	17.511	8.755	-8.755
Job Shop	la01	10.210	0.000	-10.210	0.000	0.000	0.000
Job Shop	la02	27.716	0.004	-27.712	1.527	-0.000	-1.527
Job Shop	la03	26.091	13.902	-12.189	2.848	0.000	-2.848
Job Shop	la04	0.008	0.008	0.000	-0.000	-0.000	0.000
Job Shop	orb01	33.702	31.474	-2.228	10.954	8.310	-2.644
Flow Shop	problem_10m_20j	4.470	2.117	-2.354	NA	NA	NA
Flow Shop	problem_3m_10j	0.000	0.000	0.000	0.000	0.000	0.000
Flow Shop	problem_3m_20j	0.000	0.000	0.000	0.000	0.000	0.000
Flow Shop	problem_5m_10j	0.000	0.000	0.000	0.000	0.000	0.000
Flow Shop	problem_5m_20j	0.000	0.000	0.000	0.000	0.000	0.000
Flow Shop	problem_5m_30j	0.000	0.000	0.000	0.000	0.000	0.000
Flow Shop	problem_8m_10j	0.000	0.000	0.000	0.000	0.000	0.000

9.3 Tempo

Problema	Instância	Tempo ant.	Tempo 5 min	Delta tempo	Razão tempo
Machine	inst_book	1.742	2.590	0.849	1.487
Machine	inst_n05_s01	0.025	0.030	0.005	1.183
Machine	inst_n05_s02	0.108	0.124	0.016	1.147
Machine	inst_n05_s03	0.021	0.028	0.007	1.332
Machine	inst_n07_s01	0.197	0.284	0.087	1.444
Machine	inst_n07_s02	0.780	1.163	0.383	1.492
Machine	inst_n07_s03	0.110	0.203	0.093	1.843
Machine	inst_n09_s01	0.715	1.125	0.410	1.574
Machine	inst_n09_s02	1.007	1.732	0.726	1.721
Machine	inst_n09_s03	1.006	5.300	4.295	5.269
Machine	inst_n11_s01	1.008	2.836	1.828	2.813
Machine	inst_n11_s02	1.009	18.006	16.997	17.842
Machine	inst_n11_s03	1.009	300.030	299.021	297.402
Machine	inst_n13_s01	1.012	216.476	215.464	213.984
Machine	inst_n13_s02	1.023	107.540	106.517	105.102
Machine	inst_n13_s03	1.021	8.019	6.998	7.856
Machine	inst_n15_s01	1.017	300.029	299.013	295.085
Machine	inst_n15_s02	1.017	300.038	299.021	294.881
Machine	inst_n15_s03	1.012	39.808	38.795	39.316
Machine	inst_n17_s01	1.017	300.027	299.009	294.897
Machine	inst_n17_s02	1.017	300.041	299.023	294.898
Machine	inst_n17_s03	1.020	300.063	299.043	294.109
Machine	inst_n19_s01	1.021	300.032	299.011	293.967
Machine	inst_n19_s02	1.020	300.030	299.010	294.109
Machine	inst_n19_s03	1.019	300.032	299.013	294.408
Machine	inst_n21_s01	1.023	300.035	299.012	293.230
Machine	inst_n21_s02	1.023	300.047	299.024	293.332
Machine	inst_n21_s03	1.031	300.042	299.011	291.111
Machine	inst_n24_s01	1.027	300.046	299.019	292.249
Machine	inst_n24_s02	1.032	300.048	299.017	290.866
Machine	inst_n24_s03	1.030	300.050	299.020	291.408
Job Shop	abz5	60.634	300.425	239.791	4.955
Job Shop	abz6	55.862	62.631	6.770	1.121
Job Shop	ft06	0.456	0.456	-0.001	0.999
Job Shop	ft10	60.041	300.010	239.969	4.997
Job Shop	ft20	60.132	300.095	239.962	4.991
Job Shop	la01	60.031	79.019	18.987	1.316
Job Shop	la02	60.036	265.933	205.897	4.430
Job Shop	la03	60.026	300.009	239.983	4.998
Job Shop	la04	50.766	54.134	3.367	1.066
Job Shop	orb01	60.045	300.023	239.978	4.997
Flow Shop	problem_10m_20j	60.042	300.031	239.988	4.997
Flow Shop	problem_3m_10j	0.453	0.470	0.018	1.039
Flow Shop	problem_3m_20j	0.429	0.435	0.005	1.013
Flow Shop	problem_5m_10j	0.288	0.287	-0.001	0.997
Flow Shop	problem_5m_20j	11.313	11.390	0.077	1.007
Flow Shop	problem_5m_30j	4.171	4.136	-0.035	0.992
Flow Shop	problem_8m_10j	5.039	5.059	0.021	1.004